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An analysis of the form of density dependence in a simulation model of a seasonal breeder undergoing control

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Abstract

A simple deterministic simulation model was constructed to examine compensatory mechanisms of a seasonal breeder following population reduction, with a view to determining the optimum time of year to apply such control. Data collected from studies on the European rabbit, *Oryctolagus cuniculus*, were used in the model. The compensatory mechanisms considered were a reduction in mortality, an increase in fecundity, an increase in the length of the breeding season, and a combination of reduced mortality and increased fecundity. Any degree of seasonality in either mortality or breeding led to an optimum time of year to apply a control policy, if some level of compensation occurred. In the absence of such compensation, there was no optimal time to perform population control. If such control is unbiased toward either sex then the optimum time of control is usually at the beginning of the breeding season (January). The optimum time for a male biased control policy was during the non-breeding season (November to January). Female biased control led to a more complex situation where the optimum was dependent upon the type of compensation and the degree of such compensation. This work has shown that the presence and form of compensatory mechanisms (density-dependence) may produce or adjust the optimum timing of control in populations which have either seasonal variation in mortality or fecundity. ©1997 Elsevier Science B.V. All rights reserved

Keywords: Density dependence; *Oryctolagus cuniculus*; Population control

1. Introduction

There has been a lot of theoretical work in the area of density dependent regulation of populations, and a number of laboratory and field studies have found such mechanisms. Density dependent regulation through reduced fecundity has been found in a variety of taxa, for example in aphids (e.g., Kidd, 1990) and rabbits (e.g., Thompson, 1994), through increased mortality in endoparasites (e.g., Coyne et al., 1991), barnacle geese (e.g., Larson and Forsluno, 1994), fur seals (e.g., Smith and Polacheck, 1981) and rabbits (e.g., Wood, 1980), and through both mechanisms in harp seals (e.g., Lett et al., 1981), and some deer (e.g., Putman, 1988). Such empirical studies have been used as the basis for modelling populations (Lett et al., 1981; Kidd, 1990; Hochberg and Waage, 1991), and some modelling work has apparently assumed such relationships without empirical study (Starfield and Bleloch, 1986; White and Harris, 1995).

A simple way to determine the mathematical function of these changes is to plot fecundity, or

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mortality, against density from examples in the literature, and establish a relationship by linear regression (Grant et al., 1984).

The majority of such work has concentrated on the deterioration of the population's rate of growth over time, with increasing population density. In the case of pest species, it would be more pertinent to examine the immediate change in population growth, after a reduction in population density due to control. This immediate change in population parameters is referred to as compensation to differentiate it from classical density dependence. Such pest species are often seasonal breeders, and the timing of control may also have an effect on the recovery rate. Since population size of a pest species varies cyclically throughout the year, it is the relative population size that is considered here.

This work aims to examine whether a difference between fecundity and mortality regulation can be found, and also if the mathematical form of the regulation has an effect on population recovery rates.

2. Methods

2.1. Data

The European rabbit, *Oryctolagus cuniculus*, a relatively short-living fecund r-adapted pest species, was chosen for the analysis. In wild populations this species has been recorded producing between 3 and 7 young per litter (Cowan, 1991), with annual productivity between 10 and 38 (Brambell, 1944; Gilbert et al., 1987). Annual mortality in the first year of life is between 88% and 99% (Webb, 1993; Wood, 1980), with adult mortality between 27% and 64% (Parer and Fullager, 1986; Cowan, 1987).

For the purpose of this analysis the population parameters were taken from studies of the rabbit in southern England where it is possible for rabbits to produce litters in most months of the year. The seasonal and age-related productivity, per female rabbit, is given in Table 1, which is calculated from an analysis of over 1000 does (Trout and Smith, 1995). The age of the does was calculated by the degree of lumbar fusion (Trout and Lelliot, 1992), and the productivity by the product of the number of implanted embryos and the proportion of females found to be breeding in each month.

The mortality of young rabbits is both age and seasonally dependent, with rabbits born at the start of the breeding season (December to February) having the highest survival and those born toward the end of the breeding season (July to November) having the lowest survival (Cowan, 1987). These data are calculated from a long-term capture–mark–recapture study, and are presented in Table 2. The survival rates for the first month of life, mostly prior to emergence from the warren, and thus prior to any capture and measurement, is assumed to be 0.9. Since the exact level is unknown, it is used for all months of birth, and will thus have no seasonal effect on the population growth. The sex ratio of rabbits at birth is 0.5 (Brambell, 1944; Cowan, 1991), and although there is some natural sex-dependent mortality (Cowan, 1987), it is not considered here in order to simplify the analysis.

2.2. Model

A simple non-spatial deterministic, two-sex, 25 age-class, simulation model was written, and run for

Table 1
The age related fecundity for each month of the year, as used in the model. Each figure is the number of young born per female during one month

Age in months	Time of year											
	Jan	Feb	Mar	Apr.	May	Jun	Jul	Aug	Sept	Oct	Nov	Dec
1–5	0	0	0	0	0	0	0	0	0	0	0	0
6–10	0.419	3.067	4.494	5.229	4.788	0.927	0.815	0.384	0	0	0	0
11–16	1.816	3.821	4.721	5.380	5.320	1.913	0.951	0.480	0.480	0	0	0.180
17–23	1.909	4.190	4.740	5.165	4.788	1.913	1.223	0.480	2.040	0	0	0.612
24 +	2.095	3.771	4.740	5.380	4.596	2.318	0.679	0.288	0	0	0	0.504
mean	1.676	3.712	4.740	5.220	5.098	1.690	0.480	0.412	0.380	0	0	0.280

Table 2
The age and seasonal survival rates for each month of age, as used in the model

Month of birth	Survival rate by age (months)												
	1	2	3	4	5	6	7	8	9	10	11	12	12+
Dec–Feb	0.900	0.562	0.772	0.881	0.936	0.936	0.937	0.936	0.936	0.936	0.933	0.893	0.923
Mar–Jun	0.900	0.519	0.712	0.812	0.867	0.896	0.909	0.910	0.902	0.886	0.861	0.822	0.923
Jul–Nov	0.900	0.430	0.591	0.674	0.719	0.743	0.754	0.755	0.748	0.735	0.715	0.682	0.923
Mean	0.900	0.521	0.715	0.816	0.871	0.900	0.913	0.914	0.906	0.890	0.865	0.826	0.923

two simulated years to generate an age-distributed initial population. This population was used to seed the model for each run. The model was then run for two years to determine a base line population size. The original seeded population was then run 12 times to include population control during the first year in a different month for each run, and the resultant population size 12 months later was compared with the base-line population for the same month. This resulted in 12 values of relative population size for each simulation, where each value represents the proportion of the expected population size which was attained following control in a particular month of the year. A second reason for considering relative population sizes is that it avoids the necessity for sensitivity analysis of the original data, since any changes will affect the two parts of the simulation in exactly the same manner. This will also permit the results to be considered as a general case for many fecund seasonally breeding species.

Three general forms of compensation were used; adjusting mortality, fecundity, or a combination of both. In all cases, the amount of such compensation could be varied from 0 to 1.Only one form of mortality compensation is considered. The mortality rate is proportionately reduced for each month by the ratio between current and expected population size:

$$m_{\text{new}} = m_{\text{old}}(1 - c(1 - r)), \tag{1}$$

where m_{old} is the original monthly mortality rate, m_{new} is the adjusted mortality rate, r is the population size ratio following control ($0 < r \leq 1$), and c is the level of compensation ($0 \leq c \leq 1$). Thus, when a population is reduced by 50%, the level of natural mortality in the same month is reduced by a maximum of 50% of the figure given in Table 1.

Three types of fecundity compensation are considered. Firstly, the monthly productivity is increased by a ratio similar to that above:

$$f_{\text{new}} = f_{\text{old}}(1 + 3c(1 - r)), \tag{2}$$

where f is the monthly productivity. Since fecundity is measured in absolute numbers, it was found necessary to multiply the degree of compensation by a factor of three to permit a rate of population recovery similar to that found for compensatory mortality. This permits a more direct comparison of the two types of compensation. Secondly, as a more realistic comparison, Eq. (2) is used without the multiplier, and with a maximum limit of nine offspring per female. Lastly, it was considered possible that the breeding season may lengthen as a direct result of a lowered population density. Thus, in addition to the above change, the breeding season was extended into October for all age classes, by using the productivity figure in August for September and October. This has two effects: speeding up the rate of population recovery, and producing a more marked breeding season. It is possible that the age of maturity may decrease when the population is reduced to low densities. However, this is not considered to occur to any great extent in this population, since physical maturity of the rabbit below four months of age is very unlikely. It is, however, a more likely scenario in longer lived species, where the reduction in age at maturity may be proportionately related to population density.

Only one form of combined compensation is considered, which uses adjusted forms of Eqs. (1) and (2):

$$m_{\text{new}} = m_{\text{old}}((1 - c(1 - r))/2),$$
$$f_{\text{new}} = f_{\text{old}}((1 + 3c(1 - r))/2). \tag{3}$$

These compensatory mechanisms are referred to as mortality, fecundity, seasonal, and combined, respectively.

Three different types of population control were considered. Firstly, an unbiased method which affects both sexes and all age-classes equally. Secondly, a male biased method in which 0.75 of culled animals were male, and lastly a female biased method in which 0.75 of culled animals were female. In all cases exactly the same proportion of the total population was removed by each method, and this was arbitrarily set at 50%. The last method is usually the preferred method of pest control since it should have the greatest effect on population recovery. However, to ensure the results are generally applicable there is no attempt to marry the three methods used with actual methods of rabbit control.

All of the results are expressed by plotting the proportion of the expected population (with no control) which is achieved twelve months after one control operation. Thus, if there are no compensatory mechanisms, and control reduces the population to 70%, then the observed proportion twelve months later is 0.7. If compensation permits total population recovery within 12 months then the final figure should approximate 1.0. The optimum month of the year to perform control is the month in which this proportion is at its lowest.

3. Results

Fig. 1a–d shows the level of population recovery for all four types of compensation, following an unbiased population control of 50%. The degree of compensation used varies from 0 (lowest line) to 1 (highest line) in steps of 0.2. It can be seen that if there is no compensation, there is no optimum time of year to perform unbiased control. Any level of compensation immediately produces a seasonal effect in optimising the timing of control. For mortality and seasonal compensation this occurs in January, for fecundity it occurs in November, and for combined compensation, in December. In Fig. 1a there is a second minima in June, although this is not very marked. This also occurs in the combined compensation (Fig. 1d), but is even less marked. Seasonal compensation (Fig. 1c) also results in a second minima, but this is due to the occurrence of two peaks as a result of the lengthened breeding season. This

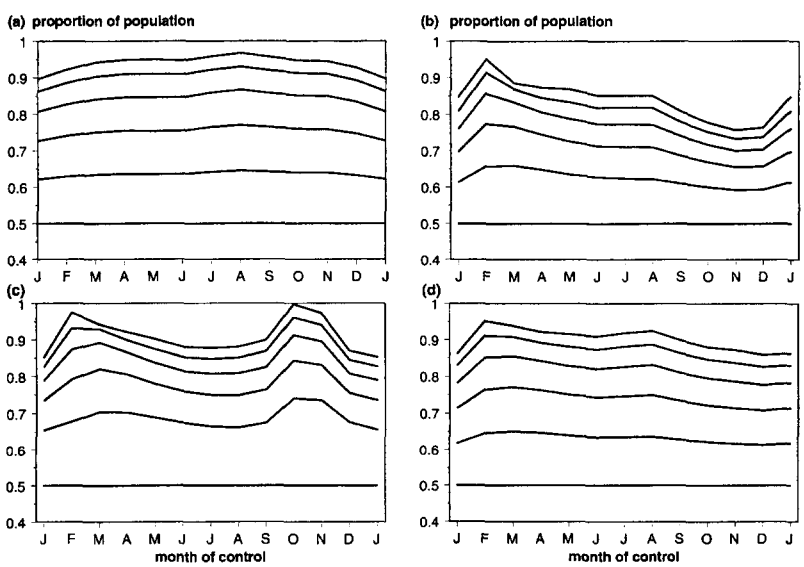

Fig. 1. The level of recovery of a population 12 months after control, as a proportion of an uncontrolled population, for control performed at different times of the year. Mortality adjusted compensation is shown in (a), fecundity adjusted compensation in (b), fecundity plus lengthened breeding season in (c) and compensation with mortality and fecundity combined in (d). The type of control is unbiased by age or sex. The degree of compensation used varies from 0 (lowest line) to 1 (highest line) in steps of 0.2.

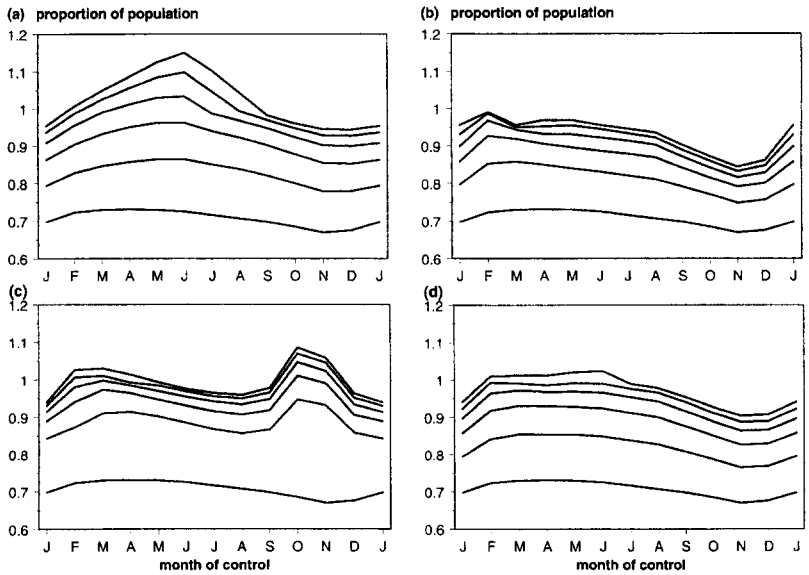

Fig. 2. As Fig. 1, but the type of control is biased toward the removal of males.

effect is seen in all of the seasonal compensation results.

If a male biased method of control is used, even a zero level of compensation results in a seasonal effect (Fig. 2a–d). Here it can be noted that with no compensation, the proportional population size is still above 0.5 at all times of year, and with a high level of compensation the population may become larger than if control had not been performed. The optimum time of year to perform control is similar, but not identical, to the unbiased method (mortality in December, seasonal in January, fecundity and combined in November).

For female biased control, a seasonal effect is noted with no compensation, the overall level of the proportional population size is below 0.5, and in no

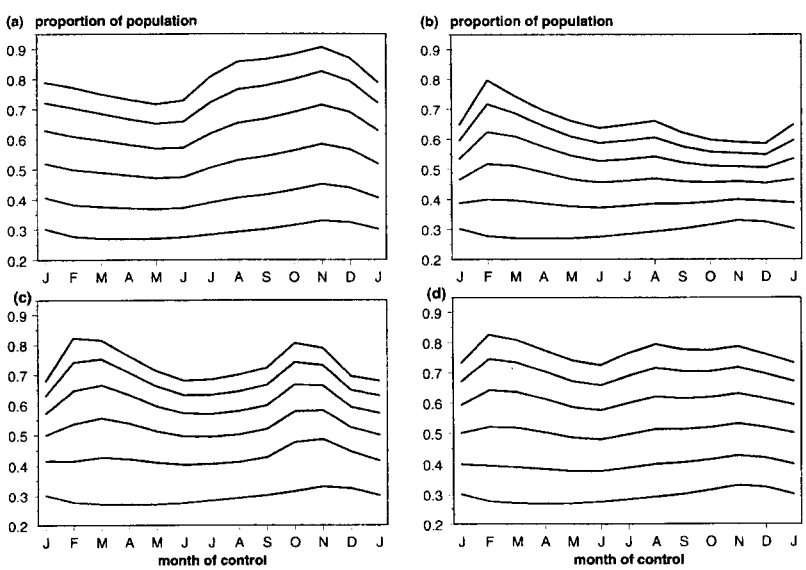

Fig. 3. As Fig. 1, but the type of control is biased toward the removal of females.

case does the population fully recover (Fig. 3a–d). Unlike the two previous methods of control, the optimum time of year is highly variable both within one method of compensation, and between methods. Without compensation this optimum is during March–May. In all cases a minima still occurs at about this time (May to June), however the optimum time shifts to December for fecundity compensation and January for seasonal compensation.

In order to determine which characteristics of the population dynamics were responsible for determining the seasonal maxima and minima, the input parameters (fecundity and mortality) were recalculated from the original data to ignore the effect of individual age and month of the year. Thus, breeding would be considered to occur throughout the year, when monthly fecundity figures were averaged. Following this, various age, or seasonally adjusted fecundity or mortality data were included until a response curve was produced that was similar to the original input data.

In a population with age related mortality, but no age-related or seasonal fecundity, no seasonal effect was found for the optimum time of control. The figures for unbiased control are given in Table 3. Including age-related fecundity did not change the annual response. These figures are presented in Table 4 for unbiased control.

If seasonal mortality, but not fecundity is now included, an optimum time of year occurs. In these cases seasonal compensation is equivalent to fecundity compensation. For unbiased control with mortality adjusted compensation, the results are shown in Fig. 4. For fecundity adjusted and combined compensation very similar results were found, but the maximum proportion attained was 0.92 and 0.94, respectively. In all three cases where compensation is used, the optimum time occurs in November. However, the seasonal fluctuation is very small when compared with Fig. 1. Fig. 5 shows the seasonal response to male biased control with mortality adjusted compensation. The results for fecundity adjusted and combined compensation were similar but with reduced seasonal variation and a maximum recovery of 0.98 and 1.00 respectively. Again the optimum time of control in all cases is during November. These graphs compare well with Fig. 2, but as for unbiased control the seasonal fluctuation is very small. Female biased control for mortality adjusted compensation is shown in Fig. 6. Here, with no compensation the optimum is in June. As in Fig. 3, mortality compensation does not affect the optimum time of year, but fecundity and combined

Table 4
The response of the population 12 months after a 50% reduction, for three types of compensation as described in the text. The population has no seasonal mortality or fecundity

Level of compensation	Type of compensation		
	mortality	fecundity	combined
0.0	0.500	0.500	0.500
0.2	0.626	0.672	0.654
0.4	0.740	0.785	0.773
0.6	0.831	0.857	0.854
0.8	0.897	0.902	0.906
1.0	0.942	0.930	0.939

Table 3
The response of the population 12 months after a 50% reduction, for three types of compensation as described in the text. The population has no age-related fecundity, or seasonal mortality or fecundity

Level of compensation	Type of compensation		
	mortality	fecundity	combined
0.0	0.500	0.500	0.500
0.2	0.631	0.681	0.661
0.4	0.748	0.799	0.785
0.6	0.841	0.871	0.867
0.8	0.908	0.916	0.919
1.0	0.951	0.944	0.950

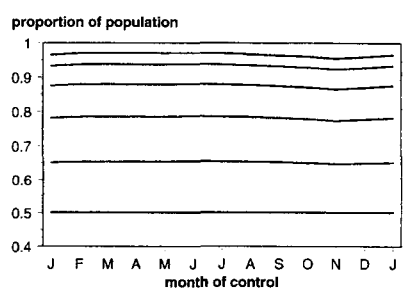

Fig. 4. As Fig. 1a. The type of control is unbiased by age or sex, and seasonal fecundity has been removed from the simulation. The results for fecundity adjusted compensation and combined compensation were similar (see text).

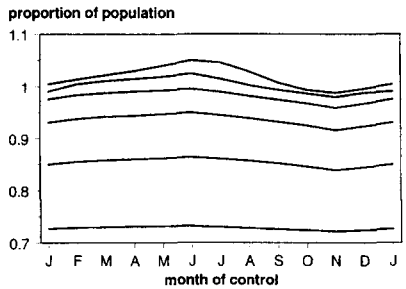

Fig. 5. As Fig. 1a. The type of control is unbiased by age, but is biased toward males, and seasonal fecundity has been removed from the simulation. The results for fecundity adjusted and combined compensation were similar but with reduced seasonal variation (see text).

compensation methods, the optimum time shifts to November with a maximum recover of 0.79 and 0.85, respectively. Again the degree of seasonal fluctuation, due to the inclusion of seasonal mortality, is very limited.

If, instead of including seasonal mortality, seasonal fecundity is included an optimum time of year again occurs. The results for unbiased control (shape of curve and the optimum timing) is very similar to the situation where the full data set is included (Fig. 1). The overall optima are found during January for mortality and seasonal compensation and in December for fecundity and combined compensation. The seasonal variation shown for mortality adjusted compensation is less than that seen in Fig. 1a, and also shows a second minima in June, which is carried over to the combined compensation. For a male

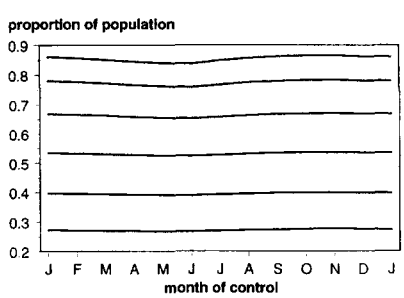

Fig. 6. As Fig. 1a. The type of control is unbiased by age, but is biased toward females, and seasonal fecundity has been removed from the simulation. The results for fecundity adjusted and combined compensation were showed similar variation but with minima in November (see text).

biased control policy the results are similar to Fig. 2, but with a reduced seasonal variation for purely mortality or fecundity adjusted compensation (a, b, and c). As above there is a great similarity between these results and the original results. Although the overall level of recovery is lower, the optima occur in December/January for mortality, December for fecundity and combined compensation, and January for seasonal compensation. For female biased control, the results are very similar to Fig. 3, with optima during May for mortality, December for fecundity, and June for seasonal and combined compensation.

4. Discussion

A deterministic sex and age class model was constructed to examine the optimum time of control for the European rabbit, and how this changes when the population can undergo different types and levels of compensation. In all cases it is assumed that 50% of the population is removed by some method of control. The practicality of this is discussed later. One assumption which was included here, and will affect the results, is the level of compensation due to fecundity. This was artificially set at three times the level of compensation due to mortality to permit a direct comparison. In the rabbit, as in most species, there is a physiological upper limit to litter size, and this should be considered in applied models.

These simulations have shown that the reason for the occurrence of an optimum time of control is the presence of either seasonal mortality rates or seasonal fecundity rates, since optima do not occur when only age related mortality and fecundity are present. This appears intuitively obvious, but there must be few species that do not undergo some degree of seasonality. In addition, if the method of control is unbiased, and no compensatory mechanisms occur, then there is no optimum time to perform control. Even with such simplistic assumptions (no seasonal fecundity or mortality) the type of compensation has an important effect on the population recovery (Tables 3 and 4). Compared to compensation due to reduced mortality, compensation as a result of increased fecundity results in a different response as the level of such compensation is in-

creased. A low level of fecundity compensation results in a greater recovery that does similar mortality compensation. At high levels of compensation this trend is reversed. A compensation which is a mathematical combination of both methods results in a level of population recovery between the previous two methods. However, at high levels of compensation, this combined method may result in a greater level of population recovery than either of the two methods individually.

When population control is unbiased toward one sex the optimum time of control is during January (near the beginning of the breeding season) except when compensation occurs by increased fecundity alone. In this situation, the optimum time of control is during November (the month immediately prior to any breeding). If only seasonal mortality is included then the optimum is always during November, whereas if only seasonal fecundity is included the optimum is usually during December or January. The exception to this is when compensation occurs through increased fecundity and a lengthening of the breeding season, which causes the optimum to move to June, although there is still a second minimum in January.

If control is biased toward males then the optimum time is during November to January regardless of the method of compensation, or which seasonal parameters are included. Despite this consistency, the general shape of the response curve in each case, is unchanged from that produced by unbiased control. Male biased control results in a greater population recovery compared to unbiased control since there is a greater proportion of females available to breed.

Female biased control reduces the level of the population by a greater extent than unbiased control because this leaves a smaller proportion of females in the population. If control is biased toward females, then the optimum time of year is highly variable, and the shape of the response curve may differ from that produced by unbiased control. In these cases it is also possible for the optimum time to shift between similar simulations with and without compensation. For example, in Fig. 3b the optimum time of control without compensation is during April, and this quickly shifts to December as the level of compensation is increased. This switch is also seen with fecundity adjusted and combined compensation. In these figures it appears that compensation by reduced mortality produces an optimum in May or June, and compensation by increased fecundity in November or December. If however, in addition to increased fecundity, the breeding season is lengthened, the optimum stays in June. Combined compensation also results in an optimum during June (Fig. 3d), and this is due more to increased fecundity than reduced mortality.

It is also important to note that all simulations involve the removal of the same proportion of the population, and not the same number of individuals, or degree of effort. Since rabbit numbers are highly seasonal, due to the high mortality of juveniles, it is likely that the optimum time of year to perform control in the field may be between November and January because of the reduced effort involved. This will be examined elsewhere.

The population dynamics of the rabbit is more sensitive to changes in the level of mortality (particularly juvenile mortality) than to changes in fecundity (Smith and Trout, 1994). These simulations support that finding; for example, comparing the mean level of recovery if seasonal fecundity is removed (Fig. 4) and if seasonal mortality is removed (effectively similar to Fig. 1). However, changes in fecundity produce a larger seasonal variation in recovery rate than do similar changes in mortality.

From the results of these simulations it can be concluded that the presence of compensation, in terms of population parameters, may produce, or adjust, the optimum time of control of a population. In addition, the type of compensation may adjust this still further, and will also have an effect on the rate of population recovery. For male biased control, it is theoretically possible to produce a larger total population as a result of such control. The result of female biased control may be much more complex than that of male biased control. Since, at least theoretically, female biased pest control is preferred, such complexity in the modelling approach should be thoroughly investigated in applied models. These types of compensation are also found in many models (e.g., Grant et al., 1984, Rodriguez, 1988, Jensen, 1995, White and Harris, 1995) which include density-dependent factors, and the exact form of such factors should also be examined during construction.

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